

SwiftRef Files – Evolution Portfolio

SSI Directory

User Guide

This document describes the data, files, and usage of the data in the SSI Directory, a new directory built in the context of the SwiftRef Files – Evolution Portfolio following the ISO 20022 migration of payments flows.

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Preface

Purpose

This document describes the data, files, and usage of the data in the SSI Directory, a new directory built in the context of the SwiftRef Portfolio Evolution following the ISO 20022 migration of payments flows.

Audience

This document is intended for anyone who installs and administers the new SwiftRef files, as well as anyone who must understand and use the new file content.

Related documentation

The following documents are related or referred to within this document and can be downloaded from the [Knowledge Centre \(User Handbook\)](#):

- *SwiftRef Portfolio Migration for ISO 20022 Migration Guides*
Current users of the existing SwiftRef files should read the Migration Guides to understand the mapping of the current files to the new files.
- *SwiftRef Release Letter* (sent by e-mail to all SwiftRef data file subscribers)
The monthly release letter announces any significant changes to the data contained in the files.
- *SSI Plus Usage Guidelines*
The in-depth guide to business aspects of how SSIs are to be created, combined, and used.

Significant Changes

The following tables list all significant changes to the content of this document since the following previous editions. These tables do not include editorial changes that Swift makes to improve the usability and comprehension of the document.

Updated information since the January 2024 edition	Location
Information about HOLIDAY SERVICES	Country Holidays Record Structure on page 9 SERVICE PMI HOLIDAY Record Structure on page 11 How to Use the SSI Directory on page 17

1 SSI Directory Content

The SSI Directory contains a set of data files that you can import into your applications or use to build your own database tables. The files enable you to find a financial institution correspondent for a currency. One file contains the SSIs while other files contain the related metadata and sets of codes (such as currency and country codes) for your convenience.

The SSI Directory contains the following files with data that is used to populate payment messages:

- RELATIONSHIPS-SSI
- CODES-RELD
- CALENDARS-CTRY
- STRUCTURES-SSID

The file name consists of the name prefix and the name suffix (for example, "CALENDARS" + "CTRY"). The prefix determines the type of data included in the file, while the suffix specifies the subset of that data.

Note *Throughout this document, the prefix and the full name are used interchangeably.*

Two additional files are included in the zip package as follows:

- EULA
This file contains a link to the terms and conditions for the usage of the data in the files and any license agreements with third-party vendors.
- sha256sum
This file contains the checksum of the zip file.

1.1 RELATIONSHIPS-SSI File

The RELATIONSHIPS-SSI file has a structure that enables for the description of a variety of relationships between institutions, each with their own record structure. Currently, the file contains the following record structures:

- **SSI**
 - Lists all Cash SSIs known by Swift.
 - Swift includes this record structure in the SSI Directory. This file is called RELATIONSHIPS-SSI.
- **PMIREL**
 - Contains the relationship between institutions of which one provides the other with access to a given PMI.
 - Swift includes this record structure in the Participants Directory. This file is called RELATIONSHIPS-PMI.

If you have subscribed to the Payments Directory, then you obtain the RELATIONSHIPS-ALL file, which combines the contents of the RELATIONSHIPS-SSI and the RELATIONSHIPS-PMI files, in one single file with different record structures.

This document describes the SSI record structure only. For more information about the PMIREL record structure, see the *SwiftRef Portfolio Evolution User Guide – For Participants Directory*.

Throughout this document, the term "entity" is applied to any type of organisational unit of a financial institution (for example, a bank):

- a head-office
- a parent office
- a department

- a domestic or foreign branch or sub-branch

Note *For a head-office, the entity may represent a legal entity. Each entity is uniquely identified by an EID. An EID is an internal 7-digit number assigned by SwiftRef to any entity listed in the directory. The EID is consistent across all SwiftRef directories.*

Throughout this document, the term "bank" is used as a synonym for a financial institution (including non-banking entities) and, in exceptional cases, corporations and government entities that own a BIC or clearing code.

See [How to Use the SSI Directory](#) on page 17 for examples about how the data in the RELATIONSHIPS-SSI file can be used.

The information in this section provides an overview of the most important data elements in the RELATIONSHIPS-SSI file. For a complete list of data attributes in the file and their definitions, see the STRUCTURES-SSID file. And within this file, see the rows where the attribute FILE NAME PREFIX is equal to "RELATIONSHIPS".

The RELATIONSHIPS-SSI file contains data regarding SSIs.

Swift's SSI offering focuses on Cash SSIs that can be used for Payments, Documentary Credits, Cash Settlements of any trade type, or in confirmation messages for over-the-counter (OTC) trades.

Terminology

For some of the fields in an SSI record, some synonyms exist as follows:

- **Account Owner** is often called "SSI Owner".
- **Account Holder** is synonymous with "Correspondent" and is often referred to as Nostro (bank).
- **Asset Categories** are also known as "Market Codes" or "Market Area Codes".

The names in **bold** in the list above are used in the RELATIONSHIPS-SSI file and the related CODES and STRUCTURES files.

In addition to the attributes that *define* the SSI (Account Owner, Currency, Asset Category, and Account Holder), each record also provides a number of metadata elements (for example, start and end-date of the SSI's validity, an optional "preferred" flag, and so on). For the two institutions (Account Owner and Account Holder), the file also contains detailed attributes such as the full names, country, and town name (city). To facilitate the centralised management as an Account Owner of all SSIs of your bank group, every SSI record also contains a code identifying the bank group the Account Owner belongs to. Similarly, to facilitate the management as an Account Holder of the SSIs of all customers of your entire bank group, every SSI record also contains a code identifying the bank group of the Account Holder.

The record structure for SSIs contains a few fields related to the **data governance** of the SSI records as follows:

- The date on which a given SSI was most recently reconfirmed by the Owner ("update date"). This can be through the Maintain Your Data environment on swiftrefdata.com, through the submission of an MT 670, or through a direct contact with Swift's data team.
- A Traffic Flag (expected in the first quarter of 2023) indicating whether Swift could find any traces in the FIN traffic of the preceding month(s) of an account relationship between the Account Owner and the Account Holder, in the currency of the SSI. A full definition of the algorithm used to determine this binary Y/N flag can be found in the *SSI Plus Usage Guidelines*.

- Two fields related to a new data governance model for SSIs that will be gradually introduced as of 2023. One field “Confirmed Inst1” will indicate whether the SSI has gone through this new procedure, from the Owner’s perspective. The second field “validated Inst2” will indicate whether the SSI was validated by the Account Holder (a new procedure, an essential part of this new Data Governance model).

Example

RECORD STRUCTURE	ATTRIBUTE 1	ATTRIBUTE 2	ATTRIBUTE 3	ATTRIBUTE 4	ATTRIBUTE 5
SSI	RELATIONSHIP TYPE	EID INST1	ROLE INST1	ID INST1	ID TYPE INST1
SSI	CORRB	1234567	Account owner	BANKBEBBXXX	BIC11
SSI	CORRB	7654321	Account owner	BBNKBEBBXXX	BIC11

ATTRIBUTE 6	ATTRIBUTE 7	ATTRIBUTE 8	ATTRIBUTE 9	ATTRIBUTE 10	ATTRIBUTE 11
COUNTRY CODE INST1	NAME INST1	TOWN NAME INST1	EID INST1	ROLE INST2	ID INST2
BE	Belgobank	Brussels	1357246	Account Holder	NOSTUS33XXX
BE	B-Bank	Brussels	1472356	Account Holder	GINKJPJTXXX

ATTRIBUTE 12	ATTRIBUTE 13	ATTRIBUTE 14	ATTRIBUTE 15	ATTRIBUTE 16	ATTRIBUTE 17
ID TYPE INST2	NAME INST2	TOWN NAME INST2	COUNTRY CODE INST2	REL USAGE ⁽¹⁾	ASSET CATEGORIES
BIC11	Nostro Bank	New York	US		ANY
BIC11	Ginkou Bank	Tokyo	JP		COPA, FOEX, MMKT

(1) **Related usage:** In future versions of the RELATIONSHIPS file, this field will list other environments where the same two parties have a relationship (for example, Swift GPI, PMI Access Provider).

ATTRIBUTE 18	ATTRIBUTE 19	ATTRIBUTE 20	ATTRIBUTE 21	ATTRIBUTE 22	ATTRIBUTE 23
CURRENCY CODE	PREFERRED	UPDATE DATE	TRAFFIC FLAG ⁽¹⁾	ACCOUNT NBR INST1	GROUP EID INST1
USD	Y	20221010	Y	10406070	BD00000001G8
JPY		20211210	Y	653-0800635	BD000000016UA

(1) This is expected in the first quarter of 2023.

ATTRIBUTE 24	ATTRIBUTE 25	ATTRIBUTE 26
GROUP EID INST2	CONFIRMED INST1 ⁽¹⁾	VALIDATED INST2 ⁽¹⁾
BD0000000TPA	Y	Y
BD00000000EC	Y	Y

(1) This is expected in the first half of 2023.

Important For SSI records regarding the Start Date and Stop Date:

The example above excludes the Record House-keeping Attributes. These are common across all records and they include the “Start Date” and “Stop Date”. For SSIs, these two fields are of particular importance, as it is a best practice for any Account Owner to inform the market of modified and deleted SSIs ahead of the activation/deletion moment.

In the case of SSI files specifically, this means that SSIs with a future Start Date and/or Stop date are included in all files you can download (full, or daily/monthly updates). SSIs with a future Start Date will be tagged with F (future) in the Record Status field (also part of the Record House-keeping Attributes). Other SSIs will be tagged with C (current) in the Record Status field.

When consulting an SSI file, it is important to do separate searches for an SSI valid now (for example, to use it in a payment message), and for SSIs valid in the future (for example, to use it in a confirmation message for a trade with a value date beyond the future activation date of that SSI, or to prepare your back-office to use it or not).

SSIs that are no longer valid are not part of any history file that can be downloaded, but the information is retained by SWIFT. If for an audit investigation, or to settle a dispute, you need to know what the valid SSIs were for a given combination of Account Owner, Currency, Asset category, and date, evidence for this can be obtained from Swift.

Important How to interpret Asset Categories combined in a list:

The RELATIONSHIPS-SSI file format will often mention, for a single SSI, several Asset Categories, combining these in a comma-separated list. In the example above, the JPY relationship between BANKBEBBXXX and GINKJPJTXXX can be used for Payments, but also for the settlement of FX or Money Market transactions.

However, as soon as there is (for that SSI relationship) a difference between its usage for one asset category versus another category, that will result in two separate lines.

For example, if for a given SSI, when used for COPA (commercial payments), account number 12345 was provided by the Account Owner and for FOEX (FX settlement), account number 45678, then two separate records will be present (one with COPA, and one with FOEX).

Similarly, if at some point a given relationship is only valid for COPA and the Account Owner submits the information that this is extended to ANYY (any asset category), then this extension beyond Commercial Payments will have its own Start Date. That too will result in two separate records.

1.2 CALENDARS-CTRY File

The CALENDARS file contains three separate record structures as follows:

- country holidays
- service PMI holidays
- publication schedule of SwiftRef files

- detailed data about all IANA time zones

1.2.1 Country Holidays Record Structure

This section contains a list of dates with various purposes that enable you to plan payment-related activities.

Financial centre (country) holidays and weekends

In the CALENDARS-ALL file, the records with the COUNTRY HOLIDAY record structure contain the list of banking holidays per country or financial centre. Receiving banks may be unable to process payment instructions that are received on days when the bank is not open for business, particularly when the processing requires manual or remedial actions. The file includes working-day and holiday data for specific countries, cities, and financial centres. No data is included for individual banks.

- Banking holidays often vary between different states, provinces, or cities within a country, so the detailed definition of banking holidays is by financial centre (for example, Chicago or Brussels) and not at the country-wide level. If all recorded financial centres in a country have the same status (for example, a holiday or weekend) on a given date, then the banking holiday status applies to the entire country. Generally, individual financial centres are listed and categorised as having a full-day, part-day, or no holiday on the date.
- Banking holidays recorded in the file are not necessarily "Public Holidays" for the town or country. A day when retailers or government departments are closed but banks are open and available to process payments are not recorded as a "holiday" in this file.
- "Part day holidays" (also referred to as "half-day holidays") are days in which working hours are reduced but not fully cancelled on the day.
- Because individual banks may choose not to observe a given holiday, this information is only indicative. Payment operators may need to source additional confirmations for important payments.

Attributes to specify financial centre holidays and weekends

Attribute	Description
DATE	The relevant date
COUNTRY CODE COUNTRY NAME	Specifies the country
COUNTRY HOLIDAY TYPE	Specifies the holiday status of the day regarding financial centre holidays/weekends. The possible values are as follows: • WKND: Weekend (non-working day) for all financial centres in the country • HLDY: Holiday (non-working day) for all financial centres in the country • PART: Part of a holiday (reduced working hours) for all financial centres in the country • SOME: Holiday/part-day holiday for some financial centres, but the status (full day/part day/no holiday) differs across different financial centres. • NONE: No weekend or holiday (that is, a normal working day) for all financial centres in the country.

- For the COUNTRY HOLIDAY TYPE status "SOME", the holiday situation varies between financial centres within the country. In this case, the additional columns "HOLIDAY CENTRES", "PART DAY HOLIDAY CENTRES", and "NO HOLIDAY CENTRES" specify the holiday status for each financial centre within the country. The columns are populated with financial centre names in comma-separated lists.

- For all COUNTRY HOLIDAY TYPE statuses (except for "SOME"), the single status applies to all financial centres within the country. In this case, the three additional financial centre name columns are not populated.
- The HOLIDAY NAME provides the name of the applicable holiday as supplementary information where available. It is only populated for holiday days, and not weekends. When the holiday name varies between financial centres, only one relevant name displays. This detail is for information only and may not always be supplied. It must not be the basis for automated processing.
- The DATE MAY VARY indicates holidays when the exact date of the holiday depends on factors that cannot be determined in advance with certainty (such as Islamic holidays). Holidays with this type of dependency contain the value "YES" in this column. For other holidays and weekends, the column remains blank.

Example

DATE	COUNTRY CODE	COUNTRY NAME	COUNTRY HOLIDAY TYPE
2022-02-26	DE	Germany	WKND
2022-02-27	DE	Germany	WKND
2022-02-28	DE	Germany	SOME

HOLIDAY CENTRES	PARTIAL HOLIDAY CENTRES	NO HOLIDAY CENTRES	HOLIDAY NAME	DATE MAY VARY
	Cologne, Düsseldorf, Essen, Saarbrücken	Berlin, Bremen, Frankfurt, Hamburg, Hanover, Kiel, Magdeburg, Munich, Stuttgart	Fasching	

Currency holidays and weekends

Payments that involve currency conversions may depend on FX markets being open for each of the currencies involved (including the USD FX market, if that is not already a sent/received currency).

Attribute	Description
DATE	The relevant date
COUNTRY CODE COUNTRY NAME	Specifies the country
COUNTRY HOLIDAY	Specifies the holiday status of the day regarding financial centre holidays/weekends. The possible values are as follows: • WKND: Weekend (non-working day) for all financial centres in the country • HLDY: Holiday (non-working day) for all financial centres in the country • PART: Part of a holiday (reduced working hours) for all financial centres in the country • SOME: Holiday/part-day holiday for some financial centres, but the status (full day/part day/no holiday) differs across different financial centres. • NONE: No weekend or holiday (that is, a normal working day) for all financial centres in the country.

Note "HOLIDAY" and "DATE MAY VARY" also apply to currency holidays.

Example (currency holidays/weekends for MYR in February 2022)

DATE	COUNTRY CODE	COUNTRY NAME	CURRENCY CODE	CURRENCY HOLIDAY TYPE
2022-04-16	MY	MALAYSIA	MYR	WKND
2022-04-17	MY	MALAYSIA	MYR	WKND
2022-04-19	MY	MALAYSIA	MYR	HLDY
2022-04-23	MY	MALAYSIA	MYR	WKND
2022-04-24	MY	MALAYSIA	MYR	WKND
2022-04-30	MY	MALAYSIA	MYR	WKND

For a complete list of data attributes and their definitions in the CALENDARS-CTRY file, see the STRUCTURES-IDENTDSTRUCTURES-SSID file where the FILE NAME PREFIX is equal to "CALENDARS".

1.2.2 SERVICE PMI HOLIDAY Record Structure

PMI holidays and weekends

In the CALENDARS file, records with the Service PMI Holiday record structure contain the list of holidays for a number of PMIs.

A country can have several PMIs, and a PMI can serve participants from several different countries (for example, the members of a currency zone). Some PMIs can process several currencies, typically a combination of the national currency (or the currency of the local currency zone) and one or more major international currencies.

Attributes to specify PMI holidays and weekends

Attribute	Description
DATE	The date when the holiday/weekend occurs.
SERVICE ID	A unique acronym for the clearing and settlement service. It is usually pre-fixed with the ISO country code or the AREA code (representing a geographical area including multiple countries) where the service is offered. A blank space separates the prefix and the remainder of the acronym For example, EMU TGT2 to identify TARGET2
SERVICE NAME	The name of the clearing and settlement service. For example, TARGET2
CURRENCY CODES	ISO Currency Codes of the currencies (in a comma-separated list) in which payments can be made on the Payment Service.
PMI HOLIDAY TYPE	A code that indicated whether the date is a weekend, holiday, or partial holiday for the Payment Service.
NOTES	Supplementary information that cannot be recorded in the structured attributes within the record. For example, this field can be used to indicate the opening hours on a partial holiday, which functionality will not be available, or to indicate the reason for the holiday.

Holidays for PMIs can be:

- Weekends (RTGSs tend to be closed during the weekend)
- Normal holidays. Non-weekend days on which the PMI cannot be used. These tend to coincide with major holidays for the related country or currency zone, but a PMI can be

- active when there is a holiday for the related country or a number of countries in the currency zone.
- Partial: this can refer to a part of the day (for example, an early closure on the day before a major holiday) or a part of the functionality of the system. When the holiday is Partial, the “NOTES” attribute provides further explanation.

The NOTES attribute can also be used to indicate the reason for a holiday such as a strike, natural disaster, or a major software intervention. Similarly, it can provide the information that a given weekend is exceptionally a working day for a given PMI.

Example records

ATTRIBUTE 1	ATTRIBUTE 2	ATTRIBUTE 3	ATTRIBUTE 4	ATTRIBUTE 5	ATTRIBUTE 6
DATE	SERVICE ID	SERVICE NAME	CURRENCY CODES	PMI HOLIDAY TYPE	NOTES
2024-11-01	AE UAEFTS	UAE Funds Transfer System	AED, EUR, USD	WKND	
2024-11-08	AE UAEFTS	UAE Funds Transfer System	AED, EUR, USD	WKND	
2024-11-02	EMU TGT2	TARGET2	EUR	WKND	
2024-11-03	EMU TGT2	TARGET2	EUR	WKND	
2024-12-24	EMU TGT2	TARGET2	EUR	HLDY	
2024-12-25	EMU TGT2	TARGET2	EUR	HLDY	

1.2.3 Publication Schedule

File Publication and Delivery on page 27 describes how the various full files and updates are products and distributed. The Publication Schedule contains records with the details about the dates of production.

1.2.4 Time Zone

The time zone record structure contains, for each IANA time zone, a calendar that indicates the UTC off-set for that time zone as it applies to the previous, current, and following two years.

For example, the fields that are specific for this UTC off-set information as follows:

TIME ZONE	COUNTRY CODE	COUNTRY NAME	TOWN NAME	ADDITIONAL TIME NAMES	CPS/AREA	START DATE	START TIME	END DATE	END TIME	UTC OFF-SET
America/ New_York	US	UNITED STATES OF AMERICA	NEW YORK	ATLANTA, BOSTON, JERSEY CITY, PHILADLPHIA, PITTSBURGH, PRINCETON, VIENNA (USA)	CT, DC, DE, FL, GA, MA, MD, ME, NC, NH, NJ, ANY, OH, PA, RI, SC, VA, VT, WV	12/3/2023	2:00:00	5/11/2023	2:00:00	-04:00

FILE NAME and VERSION

The FILE NAME PREFIX and the FILE VERSION define in which file the "SSI" record structure is used. The "SSI" record structures can be different in different versions of the CODES file, if it has evolved. Moreover, if the record structure changes, then Swift issues a new version of the RELATIONSHIPS file. Therefore, the version number is also necessary.

RECORD STRUCTURE

RECORD STRUCTURE contains the name of the record structure.

FILE HEADER ATTRIBUTE NAME and RECORD HEADER ATTRIBUTE NAME

FILE HEADER ATTRIBUTE NAME: The name of an attribute (column in a .txt file) valid for the entire file. It is the column name specified in the first record of the entire file, the file header. It is irrespective of the record structures of the sets of records included in the file.

RECORD HEADER ATTRIBUTE NAME: The name of an attribute (column) valid for a set of records with identical record structure. In a file, there can be one or multiple sets of records each with their own record structure and therefore with their own record header and RECORD HEADER ATTRIBUTE NAMES (column names).

ATTRIBUTE DESCRIPTION

ATTRIBUTE DESCRIPTION contains the definition of the attribute within the Record Header.

ATTRIBUTE FORMAT

ATTRIBUTE FORMAT specifies the data format of the data in the cell. The format consists of a number for its maximum length and an alphabetic character for the data type. The data types are as follows:

- a: alphabetic
- n: numeric
- an: a combination of alphabetic and numeric
- c: a character, including special characters from the UTF-8 character set

Note *The physical format of each cell in the file is identical. It is not restricted in length, is delimited by the TAB character, and uses the UTF-8 encoding.*

LIST OF VALUES

Many attributes in the files of the SSI Directory contain fixed values from a list, such as currency codes. All lists of values used in the SSI Directory are defined in the CODES file. The LIST OF VALUES attribute contains an expression for selecting the records in the CODES file that contain the list of values.

Examples of expressions to find the list of values of the attribute are listed in the second column of the following table:

ATTRIBUTE 5	ATTRIBUTE 8
RECORD HEADER ATTRIBUTE NAME	LIST OF VALUES
RELATIONSHIP TYPE	FILE NAME PREFIX = CODES > FILE VERSION = V1 > ATTRIBUTE 1 = RELATIONSHIP TYPE > ATTRIBUTE 2
REL USAGE	FILE NAME PREFIX = CODES > FILE VERSION = V1 > ATTRIBUTE 1 = REL USAGE > ATTRIBUTE 2
ASSET CATEGORIES	FILE NAME PREFIX = CODES > FILE VERSION = V1 > ATTRIBUTE 1 = ASSET CATEGORY > ATTRIBUTE 2

Description of the first example in the above table: To find a list of all possible values for a given ATTRIBUTE (for example, RELATIONSHIP TYPE), select all records with the indicated value for ATTRIBUTE 1 in the CODES V1 file. ATTRIBUTE 2 then contains the list of possible values.

Example (list of asset categories in the CODES V1 file)

ATTRIBUTE 1	ATTRIBUTE 2	ATTRIBUTE 3
ASSET CATEGORY	ANYY	Any Cash
ASSET CATEGORY	WHLS	Wholesale
ASSET CATEGORY	CASH	Cash
ASSET CATEGORY	COLL	Collections
ASSET CATEGORY	COMM	Commodities
ASSET CATEGORY	...	...

UNIQUE NATURAL KEY

A Unique Natural Key (UNK) is composed of a single attribute or a combination of a few attributes that uniquely identify a record in a set of records with a single record structure. UNK serves as an alternative to a record key and can be useful to set up database tables, import data from files, and construct data queries. The STRUCTURES file holds the definition of the UNK for each single record structure in the UNIQUE NATURAL KEY attribute, which contains a "Y" for the attributes that make up the UNK.

Important In the Identifiers Directory, the record key is driving the additions (MODIFICATION TYPE "A"), deletions (MODIFICATION TYPE "D"), and modifications (MODIFICATION TYPE "M"). SwiftRef has a unique natural key (UNK) to uniquely identify a record, but this key should not be linked to the modification flags.

The UNIQUE NATURAL KEY for the SSI records in the RELATIONSHIPS-ALL file

For the SSI records in the RELATIONSHIPS-ALL file, the UNK is somewhat special as it also includes the START DATE and the STOP DATE. The reason is that a current and a future version of an SSI record can exist in the file at the same time where the current record represents the active SSI that will be valid until its STOP DATE, and the future record represents the SSI that will become active later.

The UNK for the SSI record structure, according to the STRUCTURES file, includes the following attributes:

START DATE + STOP DATE + ID INST1 + ID INST2 + CURRENCY CODE + ASSET CATEGORIES

Note

Asset Categories are presented as a comma-separated list, which apart from the Asset Category code itself, nothing else differentiates a given SSI relationship between being combined with one code rather than another code, and being combined with any of the other listed Asset Category codes.

To obtain a truly unique natural key, the full list must be used. Considering the second example recorded above, the unique natural key would be BBNKBEBBXXX+GINKJPJTXXX+JPY+[COPA,FOEX, MMKT].

For example, in theory, such a concatenation of three Asset Categories can be considered as representing three records, each valid for one Asset Category. In the SSI Plus file, the approach was to store this as three separate records. Using the comma-separated list of Asset Category code enables the file to be more manageable.

ACCOUNT NBR INST1 is an optional field. It cannot be used as a natural key. Only in rare cases, two or more records are created, one for each ACCOUNT NBR INST1 for the same combination of INST1 + INST2 + CURRENCY CODE, and each time, different ASSET CATEGORIES (listed).

Consider that, in practice, a search in a RELATIONSHIPS-SSI file will not be for a unique SSI, but for all suitable SSIs using INST1 + CURRENCY CODE + a specific member of the ASSET CATEGORIES list of codes + a START DATE that is either absent or in the past + a STOP DATE that is either absent or in the future. For example, a search such as BANKBEBB, USD, COPA returns all INST2 that can be used for a commercial payment to INST1 in USD.

2 How to Use the SSI Directory

This section explains how to use the data in the SSI Directory files depending on your role and its type and the size of your organisation.

Typical data usage

The following information lists the typical data usage scenarios. To see the detailed steps that are necessary to execute the searches, see the subsequent detailed use cases.

- A bank B1 receives an international payment instruction from a corporate customer. The payment is in Currency C, the beneficiary bank is B2. Bank B1 must search where B2 can be paid for amounts in Currency C.
- A bank B1 enters in a Foreign Exchange transaction with bank B2, buying Currency C1 in exchange for Currency C2. To send a confirmation for this transaction, each of the two banks must mention the FX-related SSI of their counterparty, for the currency this counterparty is buying. These SSIs must be valid on the value date of the transaction (which can be in the future).

In addition to the usage scenarios for the data that is typical for this directory, this section also provides use cases for the data that is bundled with the key data of this directory (for example, CALENDARS data).

2.1 Search for the Data You Need

This section provides the procedures you must follow to find the data you need to populate in payment transactions, trade confirmation messages, or in a counterparty's banking profile.

Each of the following procedures is structured as follows:

[data you have] → [data you need]

2.1.1 Identifier, Currency, Asset Category → SSI

About this task

For the description of this task, it is assumed that the identifier of the beneficiary bank is known (possibly through a search in the IDENTIFIERS-ALL file).

Procedure

1. In the RELATIONSHIPS-SSI file, search for all records with INST1 as the Beneficiary Bank, ATTRIBUTE 18 as the Currency, and a correct ASSET CATEGORY in ATTRIBUTE 17.

To know which asset category is correct for your usage, consider the meaning of these codes and their hierarchy. Some are supersets of others (most importantly, ANYY is a superset of all codes). For more information, see the [SSI Plus Usage Guidelines](#).

2. If you find a single record for the criteria, then route the payment/settlement through the INST2 in that record either directly (if you have an RMA relationship with INST2) or through your own correspondent in that Currency.
3. If you find multiple records for the criteria, then you can freely choose which SSI (that is, which INST2 Correspondent) you can use. Your choice is determined by the following:
 - The presence/absence of a relationship (correspondent or RMA) between your own bank or any of your Correspondents in that currency and any of the banks listed as

INST2. If one single bank is both your correspondent for that Currency and for the Beneficiary Bank, then the payment is turned in a simple book transfer (which makes it faster and less expensive).

- Any data you have regarding the institutions (for example, KYC data, or observed cost/speed).
 - The metadata of the SSI: when it was last updated by the owner (UPDATE DATE) and does it have a TRAFFIC FLAG (and, in the future, has the SSI been confirmed by INST1 or validated by INST2 using the SSI-specific data governance).
4. If there is no resulting SSI and the Identifier for INST1 is a BIC11 that does not end in XXX, then begin your search again after replacing the Branch Code of the BIC11 (the last three characters) with XXX. To facilitate maintenance and avoid human error, it is best to separately specify SSIs for your non-XXX branches only when they are different from the SSIs that are specified for the XXX branch. Not all parties observe that best practice, however.

For example, if there is no resulting SSI for BANKBEBB123 as INST1, then repeat the search for BANKBEBBXXX.

5. Similarly (and at times to be combined with the rule above), if there is no resulting SSI and there is an Asset Category Y that subsumes the Asset Category X that you have used in your search, then replace X with Y and begin your search again.

For example, if you do not find an SSI with Asset Category COPA (commercial payments), then replace it in your search with Asset Category ANYY, as ANYY includes COPA. If there are no SSIs with Asset Category FOEX, then replace FOEX in your search with TREA (as TREA includes FOEX).

6. Check whether the SSI you intend to use is valid for the value date you need.

Note *In some instances, you must concatenate several SSIs to calculate a path between your own institution and the beneficiary bank. A full description of optimal route calculation, taking into account holiday and time zone data, observed speed, known costs, and PMI membership is beyond the scope of this document. For a basic description of how SSIs can be concatenated, see the [SSI Plus Usage Guidelines](#). You can also use the [SSI Path Constructor](#) in Bankers World Online.*

2.1.2 Date, Bank Country Code → Financial Centre Holiday/Weekend Status

About this task

As a payment operator, you want to determine if a recipient bank is expected to be closed for payment processing on a given day to know whether to expect delays when processing a payment.

Procedure

1. Filter the CALENDARS file to restrict it to rows with RECORD STRUCTURE "COUNTRY HOLIDAY".
2. Search for the COUNTRY CODE of the bank country (ISO 2-character country code).

If there are records for the code, then the file includes holiday details for that country.

3. Search for the DATE of the relevant day.

Either no records or one record should display as a result.

4. If a resulting record contains a COUNTRY HOLIDAY value of "WKND" or "HLDY", then banks are not expected to be open to process payments on that date.
5. If a resulting record contains a COUNTRY HOLIDAY value of "PART", then the bank is open to process payments on that date for reduced hours compared to a normal working day.
6. If a resulting record contains a COUNTRY HOLIDAY value of "NONE", then the bank is open to process payments on that date. This status only occurs when the day is a holiday for the local currency of the country, but not a banking holiday for financial centres in the country.
7. If the resulting record contains a COUNTRY HOLIDAY value of "SOME", then some financial centres are either closed or have reduced hours for payment processing but the situation varies between financial centres.
 - Of the financial centres listed in the "HOLIDAY CENTRES", "PART DAY HOLIDAY CENTRES", and "NO HOLIDAY CENTRES" columns, identify which is the relevant financial centre for the recipient bank.
 - If the relevant financial centre is listed in the "HOLIDAY CENTRES" column, then the banks are not open to process payments on that date (similar to the COUNTRY HOLIDAY value of "WKND" or "HLDY").
 - If the relevant financial centre is listed in the "PART DAY HOLIDAY CENTRES" column, then the bank is open to process payments on that date, but only for reduced hours compared to a normal working day (similar to the COUNTRY HOLIDAY value "PART").
 - If the relevant financial centre is listed in the "NO HOLIDAY CENTRES" column, then the holiday/weekend is not observed in that financial centre, despite being observed in other financial centres within the country.

2.1.3 Date, Currency Code → Currency Holiday/Weekend Status

About this task

As a payment operator, you may want to know whether the FX market in a given currency is open for settlement on a given day to expect delays when processing a payment that requires FX conversion that involves that currency.

Typically, FX conversion (especially in the case of larger amounts) may be possible only when the FX markets are open for each currency involved in the transaction, potentially as well as the market for other currencies (usually USD). In practice, this check may need to be completed once for each currency involved to be confident that a particular FX transaction could be processed on the specified day.

Procedure

1. Filter the CALENDARS file to restrict it to rows with RECORD STRUCTURE equal to "COUNTRY HOLIDAY".
2. Search for the LOCAL CURRENCY CODE of the currency (ISO 3-character currency code).

If there are records for the code, then the file includes holiday details for that currency. If no records are found, then the file does not contain holiday details for that currency.

The holiday details are arranged on the basis of the country, so a currency that is used locally by more than one country (such as EUR) appears against each country that contains it as the local currency. However, the same currency holiday days are recorded against

each of those countries. If you prefer unique resulting rows, then filter to include only rows for one relevant country.

3. Search for the DATE of the relevant day. Either no records or one record per country (with the same currency holiday status details) can be found.
 - If there are no resulting records, then there are no scheduled holidays or weekend days for that currency on that date.
 - If a resulting record is found with the CURRENCY HOLIDAY value of "WKND" or "HLDY", then the FX transactions that involve that currency are not possible on that day due to the relevant central bank being closed for settlements.
 - If a resulting record is found with the CURRENCY HOLIDAY value of "PART", then the FX transactions that involve that currency are possible on that day for reduced hours compared to a normal working day.
 - If a resulting record is found with the CURRENCY HOLIDAY value of "NONE", then the FX transactions that involve that currency are possible on that day, as the central bank for the currency does not have a weekend or holiday. This status occurs only when there is a financial centre holiday in the country, but the central bank is open.

2.1.4 Date, PMI Name → Holiday/Weekend Status

About this task

As a payment operator, you want to know whether a given PMI is open for settlement on a specific day.

Procedure

1. Filter the CALENDARS file (which is part of your directory bundle) to restrict it to rows with the RECORD STRUCTURE attribute equal to "SERVICE PMI HOLIDAY".
2. Further filter the result for rows that contain the name of the PMI in ATTRIBUTE 3 (SERVICE NAME).

If there are records for that PMI, then the file includes holiday details. The set of PMIs covered will initially be limited to TARGET2, as it replaces the specific holiday file for that service. More PMIs can be added in the future.

If no records are found, then the file does not provide holiday details for that PMI. The holiday details are arranged by PMI in alphabetical order, and then by date in chronological order.

3. Search for the DATE of the relevant day.
 - a. If the DATE is absent, then there are no scheduled holidays or weekend days for the PMI on that day.
 - b. If a resulting record is found with the PMI HOLIDAY TYPE value of "WKND" or "HLDY", then the PMI cannot be used on that date.
 - c. If a resulting record is found with the PMI HOLIDAY TYPE value of "PART", then the NOTES attribute provides information regarding the partial availability of the PMI.

For example, only for one of the possible currencies, or only before 12:00 local time.

3 File Management

3.1 File Naming

The files have structured names that convey significant information.

In the subsequent sections, the following example file names are used to explain the components and information contained in the file names:

- IDENTIFIERS-ALL-V1-D-2022-10-27.txt
- CALENDARS-CTRY-V1-M-2022-10-27.xml
- CODES-ISO-V1-D-2022-10-28-TXT.zip

3.1.1 File Name Prefix

The first part of the file name (for example, **CODES**-ISO-V1-D-2022-10-27.txt) specifies the type of data included in the file.

3.1.2 File Name Suffix

The text that follows the first dash ("-") in the file name (for example CODES-**ISO**-V1-D-2022-10-27.txt) specifies the subset of the data records that are included in the file as compared to the complete file (with the "-ALL" suffix).

For example, the file name CODES-ISO (available in the *Identifiers Directory* product) contains the ISO-issued codes, but other codes such as the payment purpose codes are not included. All codes are included in the CODES-ALL file, which is available in the Payments Directory product.

3.1.3 File Version

The "-V1-" in the file name (for example, CODES-ISO-**V1**-D-2022-10-27.txt) represents the first version of the file.

If the structure of the file changes due to attributes (or "tags" in .xml files) that are added or removed, or an included record structure changes, then a "-V2-" of the file is made available. The -V1- and -V2- co-exist for a period of time to provide you with sufficient time to adapt your applications.

Version changes are announced in advance in the monthly *SwiftRef Release Letter*. See the *Change Policy* section in the *SwiftRef Release Letter* for the number of months of advance notice or co-existence provided per type of change.

3.1.4 File Publication Frequency

You can order a Directory that is updated either daily or monthly. The single letter behind the version number (for example CODES-ISO-V1-**D**-2022-10-27.txt) indicates the publication frequency or whether the file content is full (all records) or only the delta (the changed records).

D: The file only contains the daily delta records. (That is, all data that has changed the day before the date in the file name.) It also contains the Record Headers and whether they have been changed.

M: The file only contains the monthly delta records. (That is, all data that has changed since the month preceding the date in the file name.) It also contains the Record Headers and whether they have been changed.

F: The file contains the full data. (That is, all data that was valid at the end of the day preceding the date in the file name.) It also contains the Record Headers.

The content of the full file is independent of the daily or monthly update cycle.

3.1.5 File Date

The date refers to the day at the end of which the snapshot of the data is taken and from which the file is generated.

For example, the file CODES-ISO-V1-D-**2022-10-27**.txt contains the changes made on 27 October 2022, and CODES-ISO-V1-F-**2022-10-27**.txt contains all data valid at the end of the day on 27 October 2022.

Note *Generally, the publication date of the file is the day after the file date.*

3.1.6 File Format

The file extension specifies the format of the file.

The data and metadata files are available in two formats. All users have access to both of the following formats (both UTF-8 encoded):

- .txt: TAB-separated text file
- .xml: .xml file

3.1.7 File Compression

To compress data volumes of the downloaded files, the individual .txt and .xml files are compressed into zip files.

For example, the CODES-ISO-V1-D-YYYY-MM-DD.txt file is compressed as follows:

- CODES-ISO-V1-D-YYYY-MM-DD-**TXT.zip**

The individual files are available through the Directories over FileAct (DOFA) service. For more information, see [File Delivery Channels](#) on page 26.

Note

3.2 File Structure

3.2.1 Headers

All files have a similar, common structure, as described below.

File Header

The first record in each file is the File Header. The first seven attributes (columns) in the File Header are identical in all files. They have standard names and contain record housekeeping attributes (for more information, see the section below). The housekeeping attributes are followed by 50 attributes with generic names ATTRIBUTE 1 to ATTRIBUTE 50.

Record Header

A file can contain one or multiple sets of records with each set of records having a different structure (which is defined by the Record Header). This adaptability enables you and Swift to group sets of data with different structures and similar purposes into a single file, which reduces the number of files necessary to contain all data of the *Identifiers Directory*. The Record Header contains the seven standard housekeeping attributes and, in addition, attributes with proprietary names that overwrite the file header's generic ATTRIBUTE names. The proprietary name represents the data included in the record set. The proprietary attributes are described in the STRUCTURES file.

Note *The terms "Record Header" and "Header Record" are synonymous and therefore they can be used interchangeably.*

Record Headers contain the attribute names and do not contain any payment data. They are included in the files and are marked with "H" in the RECORD CONTENT attribute. You can use the Record Headers to understand the data in the record set if you import the file into Microsoft Excel or similar data tools. Alternatively, you can use the Record Headers as table headers in an end-user's application. If you do not want to use the Record Headers, then you can skip them during the file import.

Record Headers are included in the full and delta files. Even if the record header has not changed, then it is included in a delta file as a modified (M) record for your convenience. If a Record Header will change, then it will have a START DATE specified and will be announced in the *SwiftRef Release Letter*. New and deleted Header Records are included in the delta files as added (A) and deleted (D) records.

Example

In the following table, the first record is the File Header. The second line is also a Record Header. The last three lines are data records.

Note *The table does not display the ID and its ID TYPE. Instead, they are listed in the IDENTIFIERS file to the left of the columns.*

RECORD STRUCTURE	RECORD CONTENT	ATTRIBUTE 13	ATTRIBUTE 14	ATTRIBUTE 15
IDENTIFIERS	H	DOMESTIC ACH ID	FIN(PLUS) BIC	FIN PLUS DN
IDENTIFIERS	D	104100000004	BKCHCNBJXXX	ou=XXX,o=BKCHCNBJ,o=swift
IDENTIFIERS	D	4111	MIZBILITXXX	ou=XXX,o=MIZBILIT,o=swift
IDENTIFIERS	D	601613	NWBKGB2L613	ou=613,o=BARCGB2L,o=swift

The STRUCTURES file contains the definitions of all Record Structures and also provides a mapping of the Record Header attributes on the File Header attributes.

3.2.2 Record House-keeping Attributes

Each record (line) in any file begins with the seven record-house-keeping attributes. These attributes have identical names in the File Header, the Record Header, and all data records in all files. This section describes each of the seven attributes.

Example

MODIFICATION TYPE	RECORD KEY	RECORD STRUCTURE	RECORD CONTENT	RECORD STATUS	START DATE	STOP DATE
A	BK0000003F00	IDENTIFIERS	H	RECORD STATUS	START DATE	STOP DATE
A	BK0000003F05	IDENTIFIERS	D	C		
A	BK0000003F06	IDENTIFIERS	D	C		
A	BK0000003F07	IDENTIFIERS	D	O		2019-11-10
A	BK0000003F08	IDENTIFIERS	D	C		
A	BK0000003F09	IDENTIFIERS	D	C		
A	BK0000003F10	IDENTIFIERS	D	C		
A	BK0000003F11	IDENTIFIERS	D	O		2017-03-01
A	BK0000003F13	IDENTIFIERS	D	O		2017-03-01

MODIFICATION TYPE

The MODIFICATION TYPE determines whether the record contains the following characteristics:

- **A**: a new record that must be added to your database
- **M**: a record that has been modified and must overwrite the data in your database
- **D**: a record that has been flagged for deletion and must be removed from your database

For more information about how to use the MODIFICATION TYPE attribute, see [File Import](#) on page 28.

Which attributes can be subject to M (that is, can be modified)?

It is important to understand which types of changes require the combination of a Deletion and an Addition (of a replacement record), and which types of changes can be modified without triggering the creation of another record.

- In the case of SSIs, INST1 (that is, the Owner of the SSI) can Modify the START Date and the STOP Date of an SSI record, and then only when in the future. Dates in the past cannot be modified. Typically, this will be used to announce the delay in the activation of a future SSI, or to change the date of a de-activation.
- The owner of an SSI can also Modify the status of the Preferred flag (that is, adding or removing one).
- By (re-)submitting an SSI, the SSI Owner also triggers an update to the Update Date (setting it to the current date).
- As Owner of the SSI record, INST1 will be able to take an action in the **Maintain Your Data** environment which triggers a modification (by Swift) in the CONFIRMED INST1 attribute (a functionality gradually introduced during 2023).
- As Correspondent, INST2 will be able to take an action, in this environment, triggering a change (by Swift) in the VALIDATED INST2 attribute (a functionality gradually introduced during 2023).
- Swift can modify a number of attributes based on factual or technical changes that do not affect the essence of the SSI record:
 - the Traffic Flag and the Rel Usage attribute
 - the EID and Group EID of the Owner and the Correspondent, the full names of these parties, and the TOWN NAME in their addresses.

RECORD KEY

The RECORD KEY is a unique number of the record across the file and across all files. The RECORD KEY value does not change over time, even if the content of the record is modified. For example, if a record contains a BIC, name, and address of the institution, if the address changes, then the RECORD KEY remains identical. You must use the RECORD KEY to match the records in the files that you import (for example, a monthly delta file) into your database with the records already stored in your database.

A RECORD KEY begins with a 2-letter prefix that represents the file name.

The following list includes the record key prefixes used in the Identifiers Directory file:

- **ID**
IDENTIFIERS-
- **ID**
IDENTIFIERS-ALL
- **CO**
CODES-ISO
- **CA**
CALENDARS-CTRY
- **ST**
STRUCTURES-IDENTD

RECORD STRUCTURE

The RECORD STRUCTURE attribute contains the name of the record structure applied to the data in that record. The record structure itself (the names of attributes within the record header and their definitions) is defined in the STRUCTURES file. For more information, see [STRUCTURES-SSID File](#) on page 13.

RECORD CONTENT

The RECORD CONTENT contains a character which indicates what the record contains, as follows:

- **D**: data
 - **H**: header information
- For each set of data records ("D") included in the file, there is one Record Header ("R").

RECORD STATUS

In the data records, the RECORD STATUS contains a character which indicates whether the record data is the following:

- **O**: old, obsolete
- **C**: current, valid, active
- **F**: future, not yet valid

Note

START DATE

In the data records, the START DATE contains the date when the record content is valid. If the START DATE is a future date, then the content is not valid yet. Except for SSIs (see below), only monthly delta files can contain a future START DATE. The daily files and full files contain data that is valid at present and do not contain and future data.

If the START DATE is in the past or is empty, then the data is already valid and can be used.

Applicable for SSIs only:

If a given SSI becomes valid in the future or is scheduled to expire on a given date in the future, then this must be communicated in advance to all parties. For SSIs, records with a future Start Date or Stop Date are communicated in both full files and the daily or monthly update files. In there, SSIs with a future Start Date will be tagged with F (future) in the RECORD STATUS field. Other SSIs will be tagged with C (current) in the RECORD STATUS field.

For the specific usage of Record Structures and Start Date in the case of SSIs, see [RELATIONSHIPS-SSI File](#) on page 5.

STOP DATE

In the data records, the STOP DATE contains the date when the record content is obsolete/inactive. If the STOP DATE is a future date, then the content is valid. If the STOP DATE is in the past, then the content is already obsolete. For SSIs, this means that the record will be removed from the full file and will appear in the daily or monthly delta file with a D (deleted). In general (*not applicable for SSIs*), monthly delta files can contain a future and a past STOP DATE. The daily files and full files contain data that is already (and still) valid.

3.3 File Integrity

All files are compressed in the zip archive. The calculated SHA256 checksum is included.

Note *The checksum does not provide file authentication.*

3.4 File Delivery Channels

You can obtain the files through three methods, either through push or pull mechanisms.

- **Manual download over the Internet ("pull")**

Log in to swiftrefdata.com and manually download the zip files (the login navigates you to login.swift.com, then to the Bankers World Online home page).

In Bankers World Online, navigate to **Download Area > Portfolio Evolution** and select the *Publication Cycle*. (*Daily* only displays the daily delta files if you have a daily updates subscription while *monthly* displays the monthly delta files if you have a monthly updates subscription and the full files that correspond with both the daily and monthly update cycles.) Select the *Product* and the *Format* and click **Download**.

The .zip file contains all files of the product. The .txt and .xml files are individual files from the .zip package. You can choose either to download. The *File date* enables you to download the files of the previous publication cycles. Ignore the tabs *Monthly Flat Files* and *Daily Flat Files*, as these contain files from the legacy products.

- **Automated download over the Internet ("pull")**

Implement an automated script that automatically logs in and downloads the zip files at a pre-configured time. For script examples and more information, see the [SWIFTRef Automated Download Service Technical Specifications](#).

- **Automated delivery through the FileAct service ("DOFA") over SwiftNet ("push")**

This service is only available for institutions connected to Swift. A subscription is required to use this delivery option. An additional fee is charged for files delivered over FileAct, based on the number of characters transmitted. You can select the individual files that you want to receive if you do not need the full set. The FileAct delivery also includes a re-push service in case you lose a file. For more information, see the [Directories Over FileAct Implementation Guidelines](#).

Most of the data is also available through APIs, which requires a separate subscription.

Related information

[SwiftRef API Technical Specifications - RESTful API - Version 1.6.0 - JSON Format](#)

[SwiftRef API Technical Specifications - RESTful API - Version 2.0.0 - JSON Format](#)

[SwiftRef API Technical Specifications - RESTful API - Version 2.0.0 - XML Format](#)

3.5 File Publication and Delivery

Daily delta files

Swift publishes the daily delta files every day.

The daily delta files available for manual and automated download are published by 02:00 GMT (at the latest).

The daily delta files distributed over FileAct (DOFA) are delivered daily between 02:00 GMT and 05:00 GMT.

The files are made available for distribution on the day after the date included in the file name.

The full files are not published every day, but are published once per month (on the date as specified in the following section).

Full files and monthly delta files

The full files and monthly delta files are published once per month on the Sunday eight days after the last Saturday of the previous month.

The files are available for manual and automated download from swiftrefdata.com as of 02:00 GMT. They are distributed over FileAct between 02:00 GMT and 10:00 GMT.

The file names contain the date of the Saturday, one day prior to the publication.

You can calculate the monthly publication dates as described above, but you also can download a complete list of the future monthly publication dates. They are listed in the CALENDARS-CTRY file in the "PUBLICATION SCHEDULE" record structure.

Daily versus monthly files

The process to import monthly delta files is simpler than the process to import daily delta files because of the low frequency. In addition, importing monthly delta files can be done manually or automatically. However, monthly delta files also have disadvantages: in between the monthly updates, the data may become outdated and result in failed payments. In addition, in the case of emergency updates, the daily delta files deliver you the updates on time, while the monthly files may cause you to wait for up to one month to receive the necessary corrections (potentially causing failed payments again).

For these reasons, SwiftRef recommends that you subscribe to the daily files.

Note *With daily updates, you do not need to import the files into your applications every day. You can save them and apply the changes in groups (for example, on a weekly basis). Daily file users also receive access to the monthly files at no extra charge.*

For more information, see [File Import](#) on page 28.

3.6 File Import

3.6.1 Import the Full Files

This section explains how to choose to import the files: with delta files, full files, or a combination of both.

If you apply a full file, then you must empty all present records from your database and add the records again. The import of a full file will re-sync your database with Swift's, correcting any inconsistencies that may be caused by the incorrect application of the delta files.

You should try to avoid full file imports if the imported data is tightly coupled with your own data and if it may be affected. SwiftRef recommends that you import the delta files by default and import the full files only in case you notice synchronisation issues and must reset the database.

3.6.2 Import the Delta Files

The size of delta files is typically small. On some days, when there are no changes, the files may be empty, though they are published anyway. Exceptionally, the delta file is large when entire sets of records must be changed. Swift announces such "mass updates" in the *SwiftRef Release Letter*.

Daily delta files

It is a good practice to import the daily delta files (except for the CALENDARS file) on the day of publication for consistency reasons, as they are cross-referenced and the updates could impact one another. Alternatively, if you must simplify your operations, then you can save the delta files together and apply them in the correct order in batches (for example, every Monday).

Note *The CALENDARS file is loosely integrated (with only the country code shared) and does not need to be applied at the same time as the other daily delta files.*

Monthly delta files

It is a good practice to import all of your monthly delta files (except for the CALENDARS file) on the same day for consistency reasons, as they are cross-referenced and the updates could impact one another.

3.6.3 Add, Modify, or Delete Records

When you import data from the file into your database, the MODIFICATION TYPE explains whether you should add, modify, or delete a corresponding record in your database based on the RECORD KEY.

Records, both the data and Header Records, in a full file only contain MODIFICATION TYPE "A". This assumes that you have emptied all records from your database and you will add all records from the full file.

Records in a delta file can contain the MODIFICATION TYPES "A", "M", and "D" meaning that records should be added, modified, or deleted from your database. The MODIFICATION TYPES also apply to the RECORD HEADER records. However, RECORD HEADER records contain "M" in order to include them in the delta file even if no modification has been made to them. RECORD HEADER records that have changed are exceptional and will also have the START DATE populated and will have been announced in the *SwiftRef Release Letter*.

3.6.4 Change Your Update Frequency

You can switch from the monthly update cycle to the daily update cycle (or conversely). The switch should occur on the monthly publication date.

Tip *The daily updates are preferable over the monthly updates.*

Three scenarios are possible for each direction of the switch. The scenario that involves the full file is recommended as it guarantees a consistent data base situation after the switch-over. The Update Scheme below represents the sequence of the files to apply (D: daily delta file, M: monthly delta file, F: full file). The file in bold contains the monthly publication date in its file name. A full month of 28, 30, or 31 days is displays below as DDDDDDDD or -----.

Switching from monthly to daily updates

Option	Description
Option 1 <i>Switching without using the full file</i>	You apply monthly updates and will switch to daily updates without using the full file in the switch process. In the month you want to process the change, apply the latest monthly delta file as usual. As of the next day, begin to apply the daily delta files. The first daily delta file is dated the day of the monthly publication date plus one. Update Scheme: ----MMDDDDDDDDDDDD
Option 2 <i>Switching by using the full file (recommended)</i>	You apply monthly updates and will switch to daily updates by using the full file in the switch process. In the month you want to process the change, do not apply the latest monthly delta file. Instead, apply the full file. As of the next day, begin to apply the daily delta files. The first daily delta file is dated the day of the monthly publication date plus one. Update Scheme: ----M-----FDDDDDDDDDDDD
Option 3 <i>Switching from full files to daily delta files</i>	You apply the full files (no delta files) monthly and will switch to daily updates. In the month you want to process the change, apply the latest full file. As of the next day, begin to apply the daily delta files. The first daily delta file is dated the day of the monthly publication date plus one. At each monthly publication date, you have the option to continue to apply the full file while applying the daily delta files on all other days, or to continue to apply the daily delta files only (and no longer apply the full file). Update Scheme: ----F-----FDDDDDDDDFDDDD or ----F-----FDDDDDDDDDDDD

Switching from daily to monthly updates

Option	Description
Option 1 <i>Switching without using the full file</i>	You apply daily updates and will switch to monthly updates without using the full file in the switch process. In the month you want to process the change, apply the latest daily delta file as usual (dated with the monthly publication date). Do not apply any updates for one month until the next monthly publication cycle. On that date, apply your first monthly delta file. Update Scheme:

Option	Description
	DDDDDDDDDDDDDD-----M----
Option 2 <i>Switching by using the full file (recommended)</i>	You apply daily updates and will switch to monthly updates by using the full file in the switch process. In the month you want to process the change, do not apply the latest daily delta file. Instead, apply the full file. Do not apply any updates for one month until the next monthly publication cycle. On that date, apply your first monthly delta file. Update Scheme: DDDDF-----M-----M----
Option 3 <i>Switching from daily delta files to full files</i>	You apply the daily delta files and will switch to monthly updates with the full file rather than the monthly delta file. In the month you want to process the change, do not apply the latest daily delta file. Instead, apply the full file. As of the next month, continue to apply the full file once per month. Update Scheme: DDDDF-----F-----F----

3.6.5 Emergency Updates

It is possible that there are significant and urgent changes that must be applied. This can be due to technical or data content mistakes made during publication or regulators that request updates about a specific date that does not coincide with the monthly publication.

In such cases, if you are a daily file user, then you are not required to do anything differently as the data is automatically updated the next day.

If you are a monthly file user, then you will for once replace the monthly delta file by 30 or 31 daily delta files. You must manually download the daily delta files that have been published since the latest monthly update and apply them in the correct order immediately. This corrects your data as soon as possible.

You must continue to apply the daily delta files until the next monthly update, and you must not apply the monthly delta file. The regular monthly delta file will also contain the emergency update and you must apply it as usual in case you decide not to follow this emergency daily update process.

4 Changes to the Data, Files, and Records

4.1 Data Updates

Generally, regular updates to the data content are considered "business as usual" and are not announced in the *SwiftRef Release Letter*. Exceptions could occur if there is a high volume of data changes or if the changes affect the data of an entire country.

4.2 Record Structure Updates

Updates to the structure of the records (and, therefore also changes to the content of the STRUCTURES file) are announced in advance in the *SwiftRef Release Letter*. If there is a record structure change, then the file containing the affected records updates its version number.

For example, if the ID TYPE record structure changes, then the CODES-ISO-V1- name updates to CODES-ISO-**V2**. Both the original and the new files are provided in parallel for a number of months, as specified in the Change Management section of the *SwiftRef Release Letter*.

4.3 File Structure Updates

It is expected that the flexibility of the record structure is sufficient to accommodate for any required future changes, and in particular, the addition of other relationship types beyond SSIs. In exceptional cases when the number of the (50) available attributes is insufficient and must be extended, the file structure (row length, number of columns) must change, the file version is updated and during an extended grace period, the original and new versions of the file are available in parallel. Additionally, this type of change is announced in advance in the *SwiftRef Release Letter*.

5 Customer Support

5.1 File Management

If you require help to handle files or to obtain access to the files or documentation, then create a support ticket. You will receive a response or will be contacted by an expert at Swift.

[Contact Swift Support.](#)

For more information, see swift.com/support.

5.2 Data Management

If you have an issue with the accuracy or completeness of the SSI Directory data content, then create a support ticket. The SwiftRef Support team will connect you with the related SwiftRef Data Collection team.

[Contact Swift Support.](#)

For more information, see swift.com/support.

Specifically for SSIs, the most appropriate way to create such a support ticket is to use the **Feedback** option in the [Bankers World Online](#) (BWO) interface on Swift's reference data (or any of the specific screens for SSIs in BWO). The behaviour of the feedback option is context-sensitive, and the data you have a question about is automatically copied into your query.

Providing your own data

For certain types of reference data, Swift can only obtain them from the parties mentioned in that data. The most important examples are Swift GPI and SSIs. For these types of reference data, Swift offers a "Maintain Your Data" environment on swift.com where authorised users can submit and manage this data.

Operators using these environments must be granted with specific roles by their swift.com Administrator. Swift recommends users to have several operators authorised to these functions and to configure in a 4-eyes control set-up. In such a set-up, the input of each operator is only effective when confirmed by a second operator. This type of 4-eyes check is optional, but highly recommended, to intercept accidental human errors as well as fraud.

When using the Maintain Your Data environment, specific assistance can be obtained through the **Feedback** option on the right side of the screen.

- [Maintain Your Data - Swift GPI](#)
- [Maintain Your Data - SSIs](#)
- [Maintain Your Data User Guide](#)
- [SSI Plus Usage Guidelines](#)

5.3 User Documentation

The following two types of user documentation are related to the SwiftRef Portfolio Evolution :

- **user guides**, such as this document
- **migration guides**, for users of the legacy products that are moving to the new Portfolio Evolution files.

Both types of user documentation can be downloaded from the swiftrefdata.com [Download Area](#) in the new *Portfolio Evolution* section.

5.4 Release Letter

The *SwiftRef Release Letter* is a document that is published monthly through e-mail. It is also downloadable through the swiftrefdata.com [Download Area](#) in the *Release Letter* section.

The *SwiftRef Release Letter* describes recent and future changes to the data and the file structure. Only major changes to the data with significant impact are announced, while minor changes are treated as "business as usual" and are not announced. Changes to the file structure are very exceptional and are announced in advance.

For a detailed description of the grace periods, read the *Change Policy* section in the *SwiftRef Release Letter*.

Appendix A Change Policy for the New Portfolio Evolution Files

This section explains how Swift handles changes to the new products of the SwiftRef Portfolio Evolution and what the impact is for users.

Change to data content	Type of change	Mandatory/ optional	Advance notice period	Impact on data file	Impact on Structures file	Impact on zip archive file	Example(s)
Small data changes or corrections Additions, modifications, or removals of records (but not entire sets)	Small data change	Mandatory	No notice	No version change	No change	No change	- correcting a BIC - adding an address - removing a PMI participant
Addition of a new data set	Medium data change	Optional	No notice	No version change	New structure added	No change	- adding a national ID set - adding a PMI participant list - adding a set of payment codes
Deletion of a data set	Large data change	Mandatory	Four months	No version change	Record structure removed	No change	- removing a national ID set - removing a PMI's participant list - removing all time zones
Addition of a record in the CODES file	Small data change	Mandatory	No notice	No version change	No change	No change	adding the value "S - Suspended" to the list of record status values that already include "O - Obsolete", "C - Current", and "F - Future"

Change to a record structure	Type of change	Mandatory/ optional	Advance notice period	Impact on data file	Impact on Structures file	Impact on zip archive file	Example(s)
Addition of a new attribute in a spare column (and new data)	Small structural change	Optional	One month	No version change	Record structure modified	No change	Adding a time zone column
Insertion of a new attribute (column) in-between existing attributes (and new data), or the removal of an attribute Note: This is avoided as much as possible.	Large structural change	Mandatory	Six months	New version added	New record structure version added	No change	Inserting the "superseding currency" in-between the currency and the country code columns
Change to the format of an existing attribute (column) (Current data not affected)	Medium structural change	Mandatory	Four months	No version change	Record structure modified	No change	Changing a country code field to a comma-separated list of country codes

Change to a record structure	Type of change	Mandatory/ optional	Advance notice period	Impact on data file	Impact on Structures file	Impact on zip archive file	Example(s)
							Changing the ID attribute length from 20 to 30 characters
Change to the name of an existing attribute (column) (Data not affected)	Medium structural change	Mandatory	Four months	No version change	Record structure modified	No change	Renaming TIME ZONE to ISO TIME ZONE
Addition of a new record structure (and data set)	Small structural change	Optional	One month	No version change	New record structure added	No change	Adding a new data set of payment rules to the CODES file
Removal of a current record structure (and data)	Medium structural change	Mandatory	Four months	No version change	Record structure removed	No change	Removing the data set of currency codes from the CODES file

Change to a zip archive file	Type of change	Mandatory/ optional	Advance notice period	Impact on data file	Impact on Structures file	Impact on zip archive file	Example(s)
Addition of a new file	Small structural change	Optional	One month	New file added	New record structure(s) added	ZIP changed	- Adding the LEI File to the SEPA Directory - Adding a new version of an existing file with 10 new columns appended to the existing 57 columns
Removal of a data file	Large structural Change	Mandatory	Six months	File removed	Record structure removed	ZIP changed	- Removing the LEI File from the Identifiers Directory - Removing an old version of an existing file

Appendix B Glossary

ACH	An Automated Clearing House for clearing low-value payments
API	Application Programming Interface A set of application-to-SwiftRef database calls for searching SwiftRef payment data automatically.
Asset Category	Market Code or Market Area Code, related to SSIs For example, ANYY, COPA, or FOEX.
Attribute	A data element The name of an attribute corresponds to the name of a column of data in a file or database.
Bank	Financial institution capable of processing customer payments (used as a synonym for financial institutions in this document)
BIC	Business Identifier Code, and identifier issued by Swift to a financial or non-financial institution The BIC can be eight characters long, or 11 characters in case it also contains the branch code. "Swift code" is a synonym. ISO Standard 9362.
BIC11	A BIC with a length of 11 characters It contains the 3-digit branch or department code. The special branch code "XXX" often represents the head-office.
BIC8	A BIC with a length of eight characters It does not contain the branch code.
BWO	Bankers World Online The online query tool to search the SwiftRef payment data manually.
Change Policy	Rules describing how and when SwiftRef announces significant changes to data in the published files
CHQ	A cheque clearing system
Clearing	The process of transmitting and reconciling low-value payments from the sending bank to the receiving bank
Clearing Code	A synonym for National ID, routing number, sort code, national clearing code (NCC)
CRM	Customer Relationships Management system
CUG	Closed User Group A group of banks or corporates that use the SwiftNet functionality.
Daily	A daily updated file
Delta	A delta file, containing the data changes made since the previous publication
Direct Participant	A bank that has an account with a payment service and interacts directly with the payment service
DN	The Distinguished Name A SwiftNet service participant ID in a special format (LDAP). A DN includes at least the BIC8 of the participant.
DOFA	Directories over FileAct A secure file delivery service over SwiftNet.
DOMETIC ACH ID	A clearing code of a bank that participates in a domestic ACH
EID	The Entity ID A unique SwiftRef proprietary identifier of an entity, used throughout the files, enabling easy cross-reference. An EID consists of seven digits.

Entity	A head-office, a branch, or a department of a bank In SwiftRef files, an Entity does not represent a legal entity.
ERP	Enterprise Resources Planning system, such as SAP TM
File Header	The top row of the file that contains the generic names of the column headers (Attributes)
FIN	Swift's old generation payment messaging system using MTs (message types)
FIN BIC	A BIC of an institution that participates in the Swift FIN service
FIN(PLUS) BIC	A BIC of a bank that participates in the SWFT FIN and FINplus service
FINplus	Swift's new generation payment messaging system based on ISO 20022
Full	A full file, containing the entire set of data
FX	Foreign Exchange
Gross Settlement	Settlement of each single customer payment, without any aggregation or netting with other transactions
Header Record	A synonym of Record Header or Record Header Record
HVPS	A high-value payment system that does not work in real-time (in the SwiftRef files)
IBAN	International Bank Account Number An account number compliant with ISO 13616.
IBAN BIC	A BIC associated/issued with an IBAN
IBAN ID	The National ID in the format as embedded in an IBAN Often it is the unchanged National ID, but it also can be just the bank ID or a National ID combined with the first four characters of the BIC.
Indirect Participant	A bank that has no account or any direct connection with the payment service, and can only use the service based on a bilateral contract with a direct participant, acting as Intermediary In some cases, the payment service will not know the indirect participants.
INST	Institution
Intermediary	A Direct Participant in a Payment Service that acts as an intermediary for an Indirect Participant
ISO 20022	A generic messaging standard based on XML that enables communities to define their own adapted messaging standards. ISO 20022 also includes a set of standard codes (the Extended Codes Set), which are included in the CODES file.
LEI	The Legal Entity Identifier of a company Companies can register an LEI with a Local Operating Unit (LOU) of the Global LEI Foundation (GLEIF). An LEI begins to be used in payments, in addition to the BICs and clearing codes.
Monthly	A monthly updated file
National ID	The domestic bank ID, optionally including the branch ID. Synonyms are: clearing code, routing number, sort code, bank ID, bank/branch ID, DOMESTIC ACH ID.
NCC	National Clearing Code A synonym for National ID.
Net Settlement	Settlement resulting from processing a batch of customer payments, for the same value date and in the same currency This can occur on a daily basis, or during regular intervals throughout the day.

Netting	The process of calculating the net difference between two banks that process a batch of customer payments
Participant ID	The identifier of a bank that participates in a payment service, such as an ACH or RTGS, or a SwiftNet service
Payment Area	A group of countries that apply identical, agreed payment rules For a list of Payment Areas, see the CODES file.
Payment Service	A synonym for PMI
PMI	Payment Market Infrastructure A clearing or settlement system, such as an RTGS or ACH. In SEPA, this is also referred to as a Clearing and Settlement Mechanism (CSM).
Portfolio Evolution	The set of SwiftRef's new data products (files, APIs, and online tools) that went live in 2022 and 2023 (including the files in this directory) in contrast to SwiftRef's legacy data products (such as Bank Directory Plus or SSI Plus)
PPC	Payment Purpose Code
Record Header	A record that contains the names of the column headers (attributes) in a set of data records, together forming a table
Record Structure	The list of attributes of a record in a defined order It also contains the mapping between the File Header and Record Header attribute names.
Regex	Regular expression A formal, computer-readable data format specification.
Release Letter	A document that describes significant changes to the data published in the files, APIs, and BWO It is published monthly and sent through e-mail to SwiftRef customers. It also is downloadable from the swiftrefdata.com Download Area in the Release Letter section.
Routing Number	Synonym for National ID
RT	Real-time or near real-time (instant, fast)
RTGS	A Real-time Gross Settlement system
RT-RPS	A Real-time Retail (low-value) Payment System (or service), including "fast" and "instant" services
SCRIPT	The set of characters used by a language An ISO 15924 standard.
SEPA	Single European Payments Area
SEPA CSM	A PMI that processes SEPA payments
Service ID	A SwiftRef proprietary identifier for a payment service, such as a PMI or a SwiftNet service
Settlement	The process of two banks that pay one other the net amount due resulting from processing customer payments
SSI	Standing Settlement Instruction
Swift	The secure, global financial messaging provider
SwiftNet	Swift network and messaging layer used for FIN, FINplus, and proprietary payment services such as PMI RTGS systems and CUG services
SwiftRef	The unit at Swift that collects, manages, and publishes payment reference data
txt	The extension of a TAB-separated text file
VAS	Value Added Service

	A service implemented on FIN (such as an RTGS) and managed by a central authority (such as a central bank).
xml	The extension of an xml file
zip	The extension of a zip file that contains one or several compressed files

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